

ELEC4631 Lecture 10: Observers, Output Feedback, and the First Method of Lyapunov

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Outline

- Output feedback
- Observers
- Separation principle: combining pole-placement + observers
- First (indirect) method of Lyapunov

Output feedback

- State feedback not always feasible.
- Some states may be physically inaccessible.
- There may be too many states to measure (impractical).

Output feedback

- Output $y(t) = Cx(t) + Du(t)$.
- It is linear combination of elements of $x(t)$ and $u(t)$.
- Often only output is accessible for feedback, not state.

State observer

- $y(t)$ can contain information about state $x(t)$.
- **Idea:** Use $y(t)$ to estimate $x(t) \rightarrow$ **observer**.

State observer

- System:

$$\dot{x}(t) = Ax(t) + Bu(t); x(0) = x_0$$

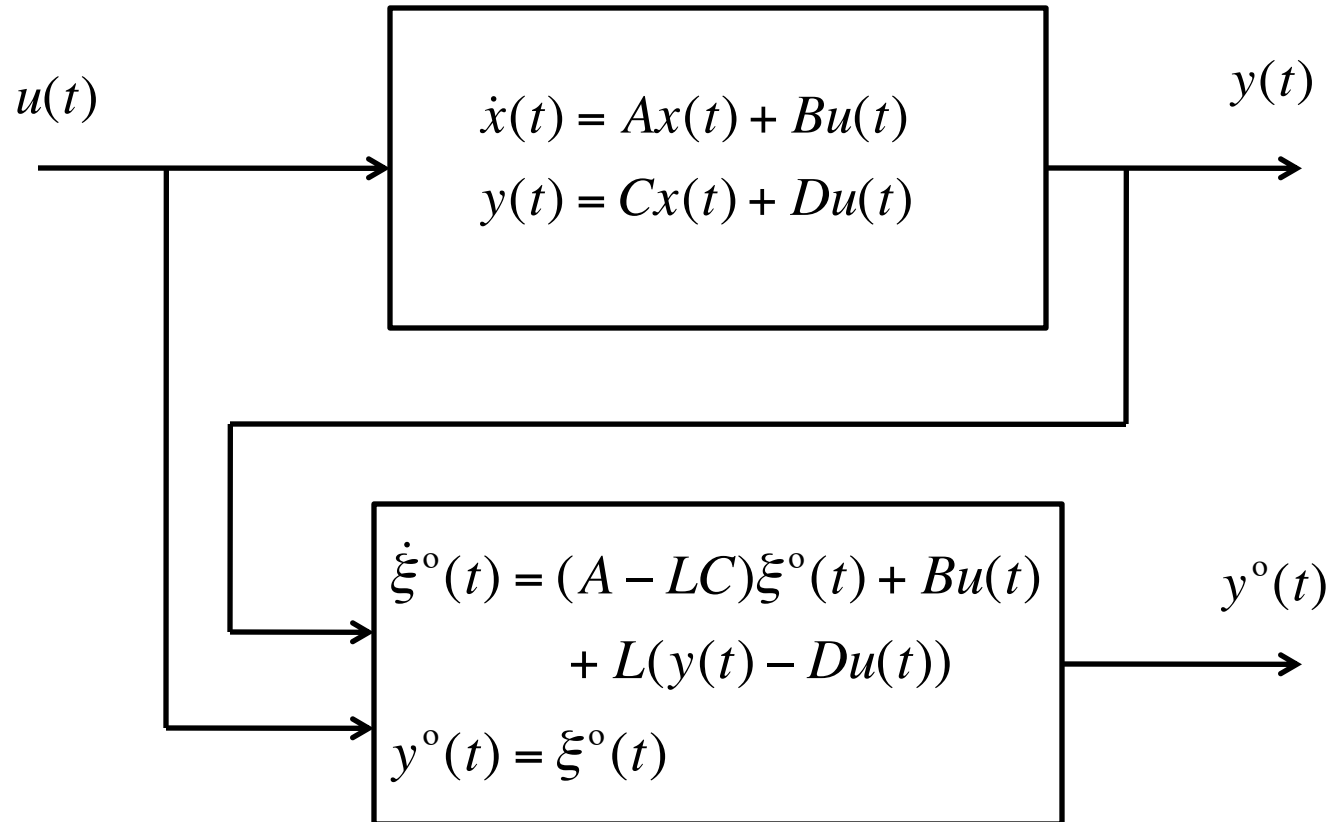
$$y(t) = Cx(t) + Du(t)$$

- Construct another system, an *observer* with state $\xi^o(t)$:

$$\begin{aligned}\dot{\xi}^o(t) &= A\xi^o(t) + Bu(t) + L(y(t) - C\xi^o(t) - Du(t)); \xi^o(0) = \xi_0^o \\ &= (A - LC)\xi^o(t) + Bu(t) + L(y(t) - Du(t))\end{aligned}$$

$$y^o(t) = \xi^o(t)$$

State observer



State observer

- Output $y^o(t)$ of observer serves as estimate of $x(t)$.
- This observer called Luenberger observer after David Luenberger (Stanford University).
- Appeared in the 60s after Kalman's state-space ideas widely adopted.

State observer

- Define error $\varepsilon(t) = x(t) - y^o(t) = x(t) - \xi^o(t)$. Then

$$\begin{aligned}\dot{\varepsilon}(t) &= \dot{x}(t) - \dot{\xi}^o(t) \\ &= Ax(t) + Bu(t) - ((A - LC)\xi^o(t) + Bu(t) - L(y(t) - Du(t))) \\ &= Ax(t) + Bu(t) - (A - LC)\xi^o(t) - Bu(t) - LCx(t) \\ &= (A - LC)(x(t) - \xi^o(t))\end{aligned}$$

- So that

$$\dot{\varepsilon}(t) = (A - LC)\varepsilon(t); \quad \varepsilon(0) = x_0 - \xi_0^o$$

and therefore

$$\varepsilon(t) = e^{(A-LC)t} (x_0 - \xi_0^o)$$

State observer

- If initial condition x_0 is known, setting $\xi^o(0) = x_0 \rightarrow \varepsilon(0) = 0 \rightarrow \varepsilon(t) = 0$ for all $t \geq 0$. Zero error = perfect estimation of state $x(t)$.
- Typically, x_0 unknown. In this case, aim to make $\varepsilon(t) \rightarrow 0$ as $t \rightarrow \infty$, as fast as allowable. Eigenvalues of $A-LC$ must be in LHP.

State observer

- L is a free column vector to play with as design parameter. So, given A and C , is there L such that $A - LC$ has eigenvalues in the LHP?
- Yes, if system is *observable*.

Controllable and observable pairs (again)

- Week 5: pair (A,B) is controllable/a controllable pair if $\mathcal{C} = [B \ AB \ \dots \ A^{n-1}B]$ is full rank.
- Week 5: the pair (C,A) is observable/an observable pair if $\mathcal{O} = [C^T \ A^T C^T \ \dots \ (A^{n-1})^T C^T]^T$ is full rank.
- Then (Week 5 too):

(A,B) is controllable/a controllable pair if and only if (B^T, A^T) is observable/an observable pair.

(C,A) is observable/an observable pair if and only if (A^T, C^T) is controllable/a controllable pair.

State observer

- Eigenvalues of $A - LC$ are same as those of $(A - LC)^T = A^T - C^T L^T$.
- If S observable $\rightarrow (C, A)$ is observable pair $\rightarrow (A^T, C^T)$ is controllable pair.

State observer

- But if (A^T, C^T) is controllable pair, from Ackermann's Formula (Lecture 6) for any desired characteristic polynomial

$$R(s) = r_0 + r_1 s + \dots + r_{n-1} s^{n-1} + s^n$$

there is row vector K such that $A^T - C^T K$ has $R(s)$ as characteristic polynomial.

State observer

- Direct application of Ackermann's Formula gives

$$K = [0 \quad 0 \quad \dots \quad 0 \quad 1](\mathcal{O}^T)^{-1}R(A^T).$$

- Setting $L = K^T$

$$L = R(A)\mathcal{O}^{-1}[0 \quad 0 \quad \dots \quad 0 \quad 1]^T$$

then characteristic polynomial of $A-LC$ is $R(s)$.

Example - observer

- Take servomotor with velocity feedback from Lecture 9 with position as output:

$$\dot{\xi}(t) = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \xi(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t); \quad \xi(0) = \xi_0$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \xi(t)$$

- Use $y(t)$ to estimate $\xi(t)$ using Luenberger observer.

Example - observer

- For servomotor, observability matrix is

$$\mathcal{O} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- $\mathcal{O}$ is invertible, so system observable. Observer can be constructed.

Example - observer

- Suppose we want eigenvalues of $A-LC$ to be $-3+4j$ and $-3-4j$.
- Then $R(s) = (s+3-4j)(s+3+4j) = s^2 + 6s + 25$.

$$\begin{aligned} R(A) &= A^2 + 6A + 25I \\ &= \begin{bmatrix} 25 & 5 \\ 0 & 20 \end{bmatrix} \end{aligned}$$

Example - observer

- Therefore,

$$\begin{aligned} L &= R(A)\mathcal{O}^{-1}[0 \quad 1]^T \\ &= [5 \quad 20]^T \end{aligned}$$

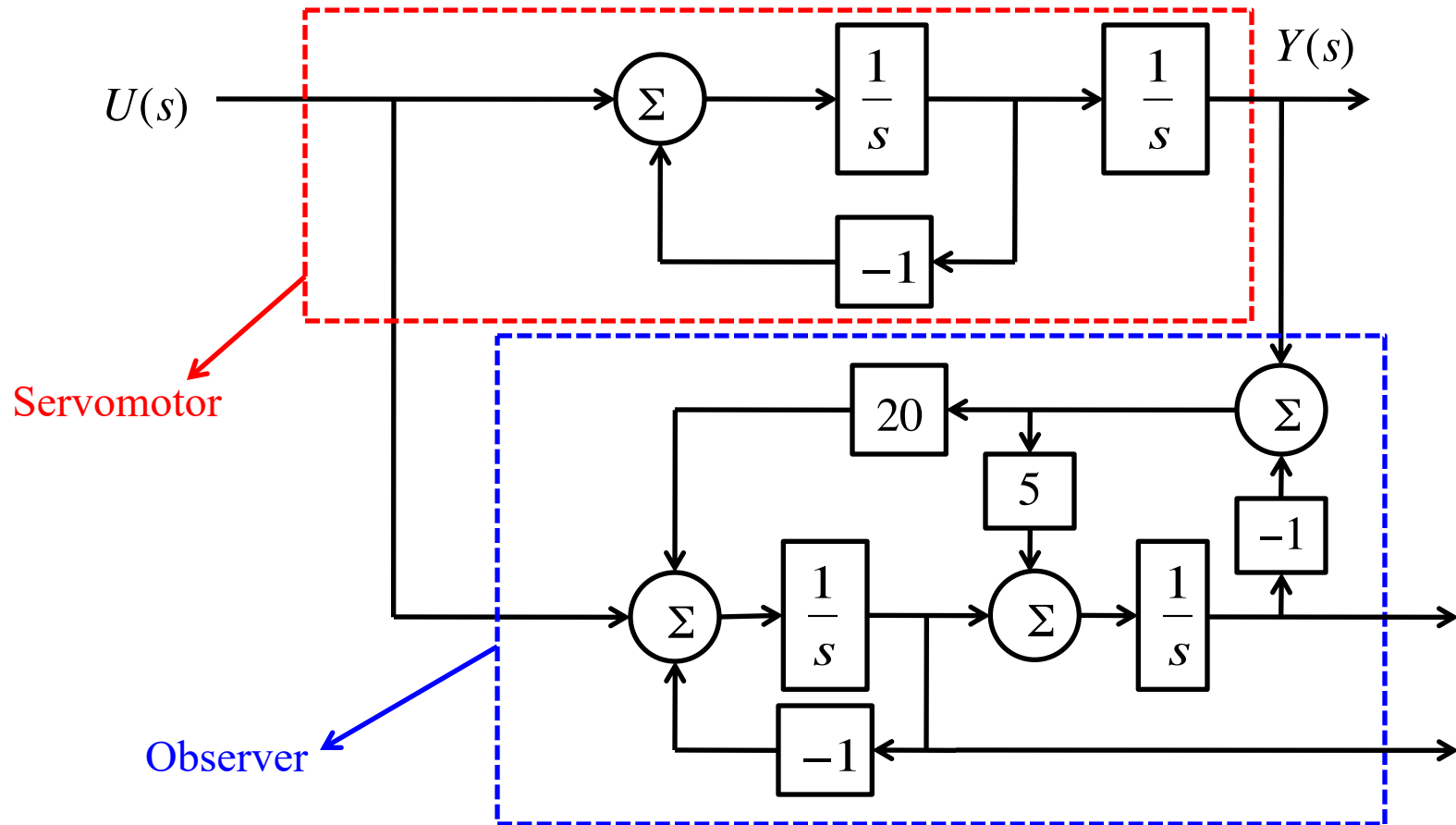
- Observer is

$$\dot{\xi}^o(t) = (A - LC)\xi^o(t) + Bu(t) + Ly(t); \quad \xi^o(0) = \xi_0^o$$

$$= \begin{bmatrix} -5 & 1 \\ -20 & -1 \end{bmatrix} \xi^o(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t) + \begin{bmatrix} 5 \\ 20 \end{bmatrix} y(t)$$

$$y^o(t) = \xi^o(t)$$

Example – observer

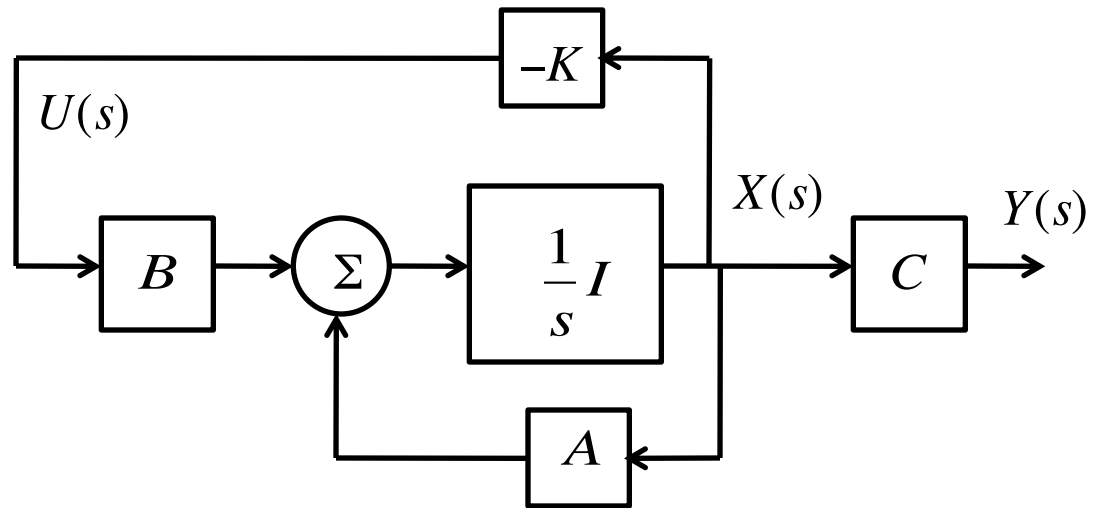
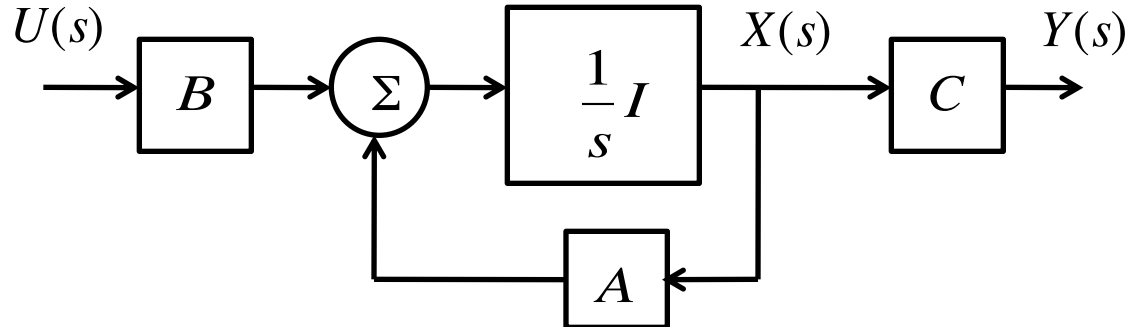


State feedback

- $\dot{x}(t) = Ax(t) + Bu(t)$
 $y(t) = Cx(t)$

(Without loss of generality, temporarily assume $D = 0$)

- If system controllable then K can be chosen such that $A - BK$ has desired eigenvalues.



Output feedback with observer

- Since state inaccessible, cannot do state feedback $u(t) = -Kx(t)$.
- However, have observer output $y^o(t) = \xi^o(t)$ available as estimate of $x(t)$.
- **Idea:** Replace $x(t)$ with its estimate $y^o(t)$ and use control signal $u(t) = -Ky^o(t)$ (output feedback).

System + Observer

- **System:**

$$\dot{x}(t) = Ax(t) + Bu(t)$$

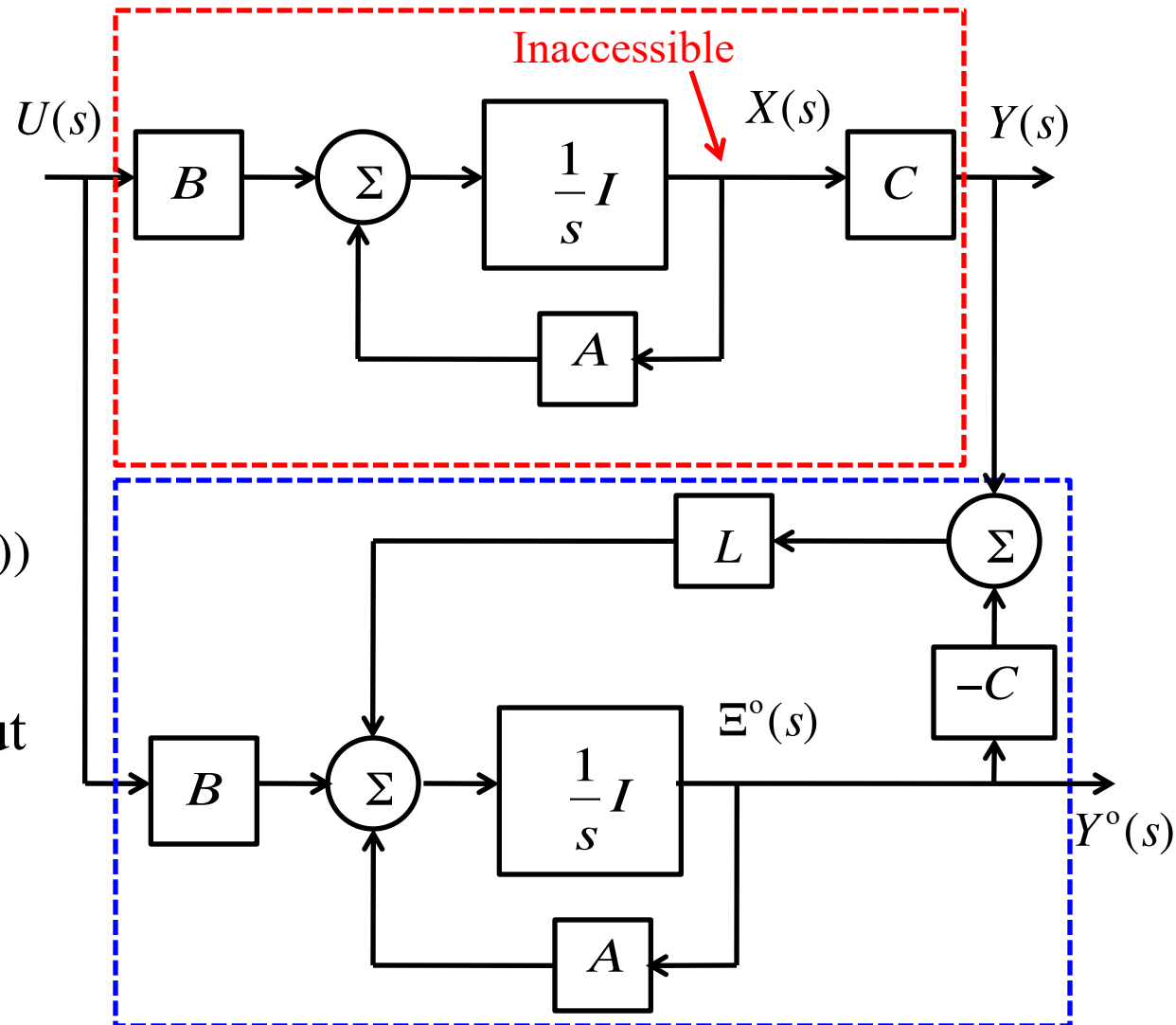
$$y(t) = Cx(t)$$

- **Observer:**

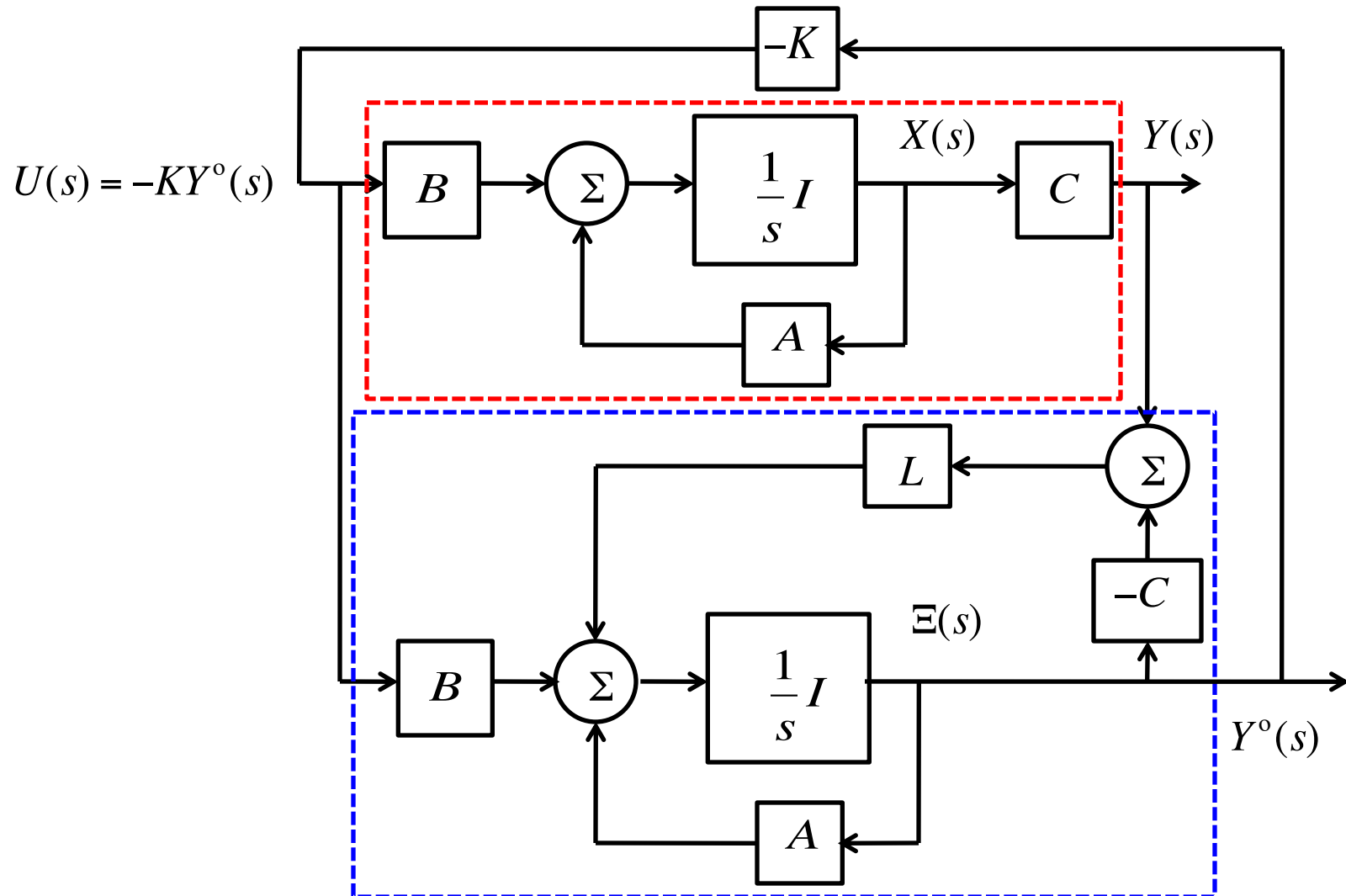
$$\begin{aligned} \dot{\xi}^o(t) = & A\xi^o(t) + Bu(t) \\ & + L(y(t) - C\xi^o(t)) \end{aligned}$$

$$y^o(t) = \xi^o(t)$$

- Cannot access $x(t)$, but do have access to $y^o(t) = \xi^o(t)$. How to control?



System + observer + feedback



Separation principle

- Sounds reasonable. But does this work? Need to analyse it.
- With $u(t) = -K\xi^o(t)$ closed-loop state-space equation for combined system + observer + feedback is:

$$\begin{bmatrix} \dot{x}(t) \\ \dot{\xi}^o(t) \end{bmatrix} = \begin{bmatrix} A & -BK \\ LC & A - BK - LC \end{bmatrix} \begin{bmatrix} x(t) \\ \xi^o(t) \end{bmatrix}$$

Separation principle

- Let

$$A_{\text{cl}} = \begin{bmatrix} A & -BK \\ LC & A - BK - LC \end{bmatrix}$$

and define orthogonal matrix

$$W = \frac{1}{\sqrt{2}} \begin{bmatrix} I_{n \times n} & I_{n \times n} \\ I_{n \times n} & -I_{n \times n} \end{bmatrix}$$

(i.e., $W^{-1} = W^T$)

Separation principle

- Then

$$\begin{aligned} W^{-1}A_{\text{cl}}W &= \frac{1}{\sqrt{2}} \begin{bmatrix} I_{n \times n} & I_{n \times n} \\ I_{n \times n} & -I_{n \times n} \end{bmatrix} \begin{bmatrix} A & -BK \\ LC & A - BK - LC \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} I_{n \times n} & I_{n \times n} \\ I_{n \times n} & -I_{n \times n} \end{bmatrix} \\ &= \begin{bmatrix} A - BK & BK + LC \\ 0 & A - LC \end{bmatrix} \end{aligned}$$

Separation principle

- Since $W^{-1}A_{cl}W$ is similar to A_{cl} it has same characteristic polynomial:

$$\begin{aligned}\det(sI_{2n \times 2n} - A_{cl}) &= \det(sI_{2n \times 2n} - W^{-1}A_{cl}W) \\ &= \det\left(\begin{bmatrix} sI_{n \times n} - (A - BK) & -BK - LC \\ 0_{n \times n} & sI_{n \times n} - (A - LC) \end{bmatrix}\right) \\ &= \det(sI_{n \times n} - (A - BK))\det(sI_{n \times n} - (A - LC))\end{aligned}$$

Separation principle

$$\det(sI_{2n \times 2n} - A_{cl}) = \det(sI_{n \times n} - (A - BK)) \times \det(sI_{n \times n} - (A - LC))$$

Characteristic polynomial of the system alone with state feedback gain K (is independent of L)

Characteristic polynomial of the observer alone with observer gain L (is independent of K)

- Closed-loop system has as eigenvalues the combined eigenvalues of (i) the system alone under state feedback gain K , and (ii) the observer alone with observer gain L .

Separation principle

$$\det(sI_{2n \times 2n} - A_{cl}) = \det(sI_{n \times n} - (A - BK)) \times \det(sI_{n \times n} - (A - LC))$$

Characteristic polynomial of the system alone with state feedback gain K (is independent of L)

Characteristic polynomial of the observer alone with observer gain L (is independent of K)

- **Separation principle** (of linear output feedback control):

State feedback gain K and observer gain L can be designed separately and independently of one another. Then combined in control system.

Separation principle

- **Keep in mind:** separation principle requires for system to be controllable (so feedback gain K exists) and observable (so observer gain L exists).
- Controllability and observability conditions can be relaxed but not covered here.

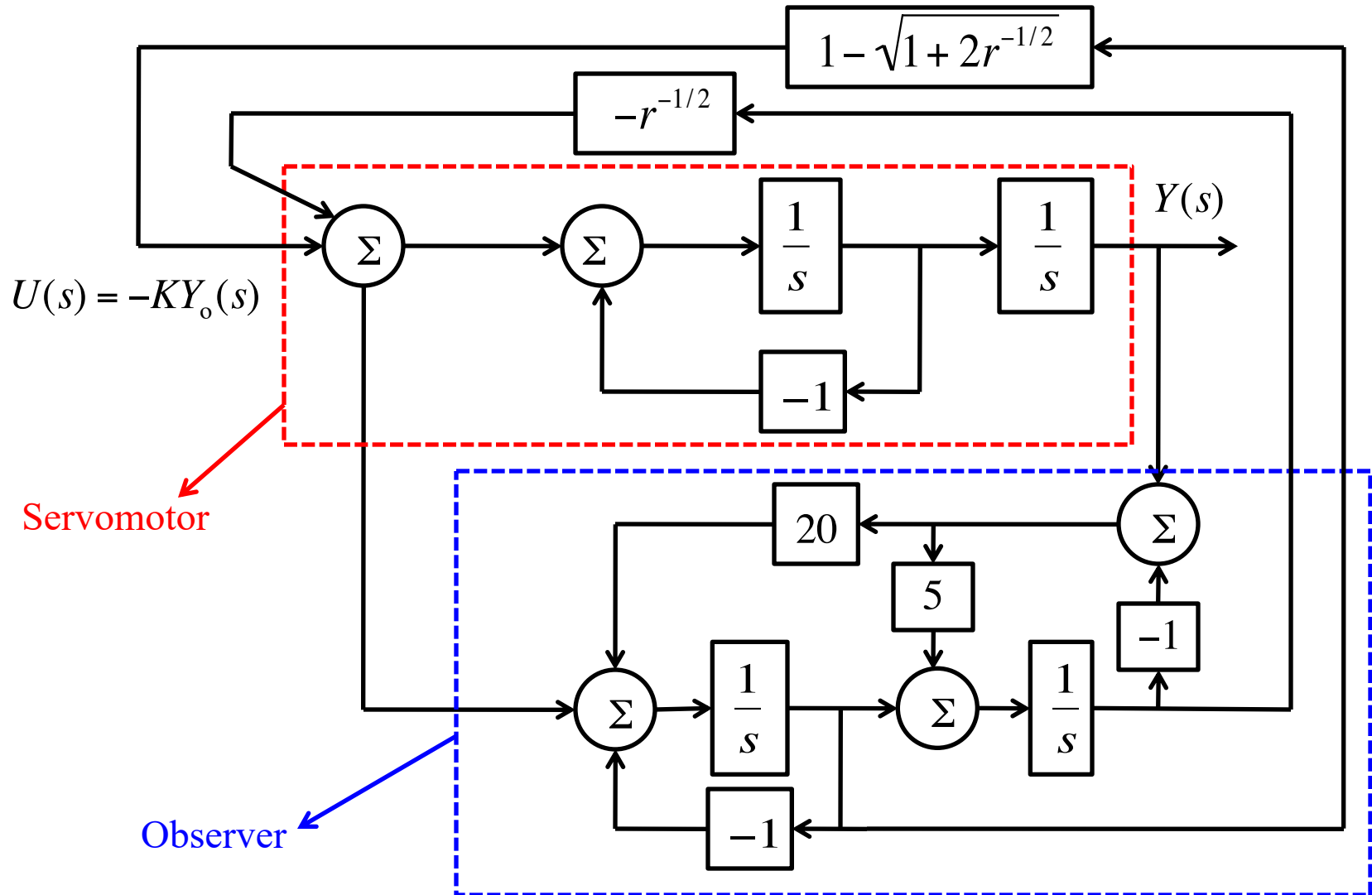
Example – separation principle

- For servomotor with velocity feedback, feedback gain K has been designed in Lecture 7 with LQR theory ($r > 0$):

$$K = \begin{bmatrix} r^{-1/2} & \sqrt{1 + 2r^{-1/2}} - 1 \end{bmatrix}$$

- This lecture, observer has been separately designed for servomotor.
- They can be combined to form control system.

Example – separation principle



The case $D \neq 0$

- **System:**

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

- **Observer:**

$$\dot{\xi}^o(t) = A\xi^o(t) + Bu(t) + L(y(t) - C\xi^o(t) - Du(t))$$

$$= A\xi^o(t) + Bu(t) + L(Cx(t) + Du(t) - C\xi^o(t) - Du(t))$$

$$= A\xi^o(t) + Bu(t) + L(Cx(t) - C\xi^o(t))$$

$$y^o(t) = \xi^o(t)$$

- D plays no role in determining K .

The case $D \neq 0$

- Observer:

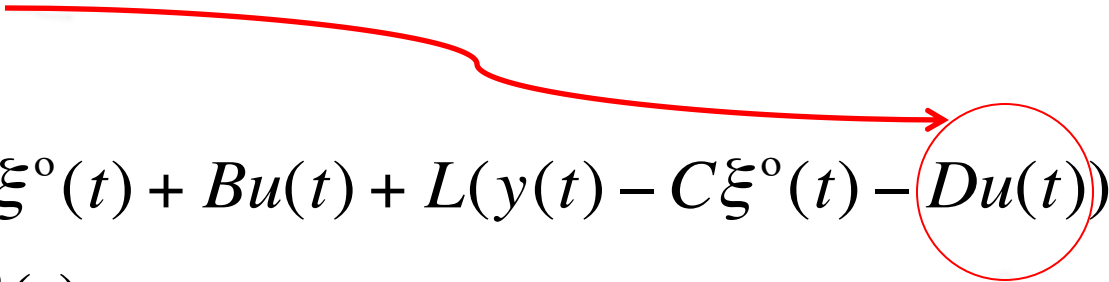
$$\begin{aligned}\dot{\xi}^o(t) &= A\xi^o(t) + Bu(t) + L(y(t) - C\xi^o(t) - Du(t)) \\ &= A\xi^o(t) + Bu(t) + L(Cx(t) + Du(t) - C\xi^o(t) - Du(t)) \\ &= A\xi^o(t) + Bu(t) + LC(x(t) - \xi^o(t))\end{aligned}$$

$$y^o(t) = \xi^o(t)$$

- D taken into account in observer by the term $L(y(t) - C\xi^o(t) - Du(t))$ but it gets cancelled, so it also plays no role in determining L .

The case $D \neq 0$

- K, L can be designed as if $D = 0$.
- But if $D \neq 0$, D needs to appear in the observer equations.

$$\dot{\xi}^o(t) = A\xi^o(t) + Bu(t) + L(y(t) - C\xi^o(t) - Du(t))$$


$$y^o(t) = \xi^o(t)$$

First/indirect method of Lyapunov

- Stability analysis of general ODE

$$\dot{x}(t) = f(x(t)), \quad f(x) = \begin{bmatrix} f_1(x) \\ \vdots \\ f_n(x) \end{bmatrix}$$

not easy.

- Know already one approach: direct or second method of Lyapunov using Lyapunov functions (Lecture 3).

First/indirect method of Lyapunov

- Suppose x^* is an equilibrium point ($f(x^*) = 0$) and f has Taylor series expansion around x^* :

$$f(x) = \underbrace{f(x^*)}_{=0} + \text{Jac}_{x^*}(f)(x - x^*) + \text{higher order terms}$$

when $\|x - x^*\| < r$, some small $r > 0$.

- If $z(t) = x(t) - x^*$ then

$$\dot{z}(t) = \text{Jac}_{x^*}(f)z(t) + \text{higher order terms}$$

First/indirect method of Lyapunov

- If $z(0)$ close enough to x^* then approximate:

$$\dot{z}(t) \approx \text{Jac}_{x^*}(f)z(t)$$

- $\text{Jac}_x(f)$ denotes Jacobian matrix of f at the point x :

$$\text{Jac}_x(f) = \begin{bmatrix} \frac{\partial f_1(x)}{\partial x_1} & \frac{\partial f_1(x)}{\partial x_2} & \dots & \frac{\partial f_1(x)}{\partial x_n} \\ \frac{\partial f_2(x)}{\partial x_1} & \frac{\partial f_2(x)}{\partial x_2} & \dots & \frac{\partial f_2(x)}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n(x)}{\partial x_1} & \frac{\partial f_n(x)}{\partial x_2} & \dots & \frac{\partial f_n(x)}{\partial x_n} \end{bmatrix}$$

First/indirect method of Lyapunov

- First method of Lyapunov: checking stability of x^* by inspecting eigenvalues of $\text{Jac}_{x^*}(f)$.

First/indirect method of Lyapunov

- If $\text{Jac}_{x^*}(f)$ has all its eigenvalues in the LHP $\rightarrow x^*$ is asymptotically stable.
- If $\text{Jac}_{x^*}(f)$ has *at least one* eigenvalue in RHP $\rightarrow x^*$ is unstable. Unstable means x^* not stable i.s.L.

First/indirect method of Lyapunov

- If $\text{Jac}_{x^*}(f)$ has eigenvalues on imaginary axis and all other eigenvalues are in LHP (thus no eigenvalues on RHP), no conclusion can be drawn by this method. A more sophisticated analysis is required to study stability of x^* (center manifold theory--not covered here).
- If there is only one equilibrium point, the indirect method also cannot be used to ascertain whether this point is globally asymptotically stable.

Example – First method of Lyapunov

- System ODE is

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -2x_1(t) + x_1(t)x_2(t) \\ -x_2(t) + x_1(t)x_2(t) \end{bmatrix}, \quad f(x) = \begin{bmatrix} -2x_1 + x_1x_2 \\ -x_2 + x_1x_2 \end{bmatrix}$$

- Equilibrium points are (two of them):

$$f(x) = \begin{bmatrix} -2x_1 + x_1x_2 \\ -x_2 + x_1x_2 \end{bmatrix} = 0 \Rightarrow x^* = (0,0), x^* = (1,2)$$

Example – First method of Lyapunov

- First equilibrium point $x^* = (0,0)$

$$\text{Jac}_{(0,0)}(f) = \left[\begin{array}{cc} -2 + x_2 & x_1 \\ x_2 & -1 + x_1 \end{array} \right] \bigg|_{(x_1, x_2) = (0,0)} = \left[\begin{array}{cc} -2 & 0 \\ 0 & -1 \end{array} \right]$$

- Jacobian has eigenvalues -2 and -1 (all in LHP) $\rightarrow$ $(0,0)$ is asymptotically stable equilibrium point.

Example – First method of Lyapunov

- Second equilibrium point $x^* = (1,2)$

$$\text{Jac}_{(1,2)}(f) = \left[\begin{array}{cc} -2 + x_2 & x_1 \\ x_2 & -1 + x_1 \end{array} \right] \bigg|_{(x_1, x_2) = (1, 2)} = \left[\begin{array}{cc} 0 & 1 \\ 2 & 0 \end{array} \right]$$

- Jacobian has eigenvalues $\pm \sqrt{2}$, one eigenvalue in RHP $\rightarrow x^* = (1,2)$ is unstable equilibrium point.

Example – First method cannot give conclusion

- Consider the 1-state ODE

$$\dot{x} = -x^3$$

- ODE has only one equilibrium $x^* = 0$. Equilibrium can be shown to be globally asymptotically stable with Lyapunov function $V(x) = \frac{1}{2} x^2$.
- However, Jacobian of vector field $f(x) = -x^3$ at x^* is $\text{Jac}_{x^*}(f) = 0$. So, nothing can be concluded about the stability or instability of equilibrium point using indirect/first method of Lyapunov.